A missing boundary in the Suzuki subfield construction

Human Compression project

Abstract

The number of distinct prime divisors of a field degree does not determine whether its smallest selected subfield is trivial. A repeated prime factor allows all distinct primes to be selected while leaving a simple Suzuki group at the bottom. This yields a rank-four residually weakly primitive pregeometry omitted from the eleven literal constructions in the stated classification. We give the complete subgroup argument, reusing the earlier project proof of this boundary. We do not assert a corrected exhaustive classification or a new solution of an open problem.

1 The subgroup condition

Let G be a group, let I be a finite type set, and let $G_i \le G$ for $i \in I$. The coset pregeometry has elements gG_i, of type i; two elements are incident when their cosets intersect. Put

$$G_J = \bigcap_{j \in J} G_j \quad (J \neq \varnothing), \qquad G_\varnothing = G.$$

The source calls this system residually weakly primitive when, for every $J \subsetneq I$, at least one $i \in I \setminus J$ makes $G_{J\cup\{i\}}$ a maximal proper subgroup of G_J. This is the definition on printed page 4 of the source. Its principal classification concerns pregeometries with precisely this subgroup condition; flag transitivity and the geometry axiom are not additional hypotheses of that classification.

Write $S_d = \mathrm{Sz}(2^d)$ for odd d. We use standard simultaneous subfield embeddings into the faithful four-dimensional Suzuki model. Two classical facts are retained: S_d is nonabelian simple for $d \ge 3$, and S_d is maximal in S_e if e/d is prime and $d > 1$. These are among the Suzuki structural inputs used by the source itself (its Table 1, printed page 6).

We also record why subfield intersections are exact in these embeddings. In characteristic two the standard root and diagonal matrices are

$$U(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^{\theta+1}+b & a^\theta & 1 & 0 \\ a^{\theta+2}+ab+b^\theta & b & a & 1 \end{pmatrix}, \qquad h(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}),$$

where $\theta = 2^{(e+1)/2}$ in $\mathbb{F}_{2^e}$; let w be the antidiagonal identity matrix. The standard Bruhat forms are $U(a,b)h(t)$ and $U(a,b)h(t)wU(c,f)$. This matrix description is retained classical Suzuki theory. For odd $d \mid e$, the exponent θ restricts to $2^{(d+1)/2}$ on $\mathbb{F}_{2^d}$.

Lemma 1. *For odd divisors a, b of an odd integer n, the standard embedded subgroups satisfy*

$$S_a \cap S_b = S_{\gcd(a,b)}.$$

Proof. First, the elements of S_n whose matrix entries lie in $\mathbb{F}_{2^d}$ are exactly S_d. In the first Bruhat form, the entry g_{22} gives t, the ratio g_{21}/g_{11} gives a, and g_{42}/t gives b. Thus all parameters lie in the smaller field when the entries do. In the second form,

$$g_{14} = t^{\theta+1}, \quad c = g_{13}/g_{14}, \quad f = g_{12}/g_{14}, \quad a = g_{24}/g_{14}, \quad b = g_{34}/g_{14} + a^{\theta+1}.$$

The power $\theta+1$ is coprime to 2^n-1: an odd common divisor would give both $2^{n+1}=1$ and $2^n=1$ modulo that divisor. Consequently t is a fixed integer power of g_{14}, and hence also lies in the smaller field. The two forms are distinguished by whether g_{14} is zero. This proves the asserted equality with the matrices over $\mathbb{F}_{2^d}$. Finally $\mathbb{F}_{2^a}\cap\mathbb{F}_{2^b}=\mathbb{F}_{2^{\gcd(a,b)}}$, which gives the result. □

2 The omitted pregeometry

Let $p_1,\dots,p_k$ be distinct prime divisors of an odd integer n, and put

$$d=\frac{n}{p_1\cdots p_k},\qquad G=S_n,\qquad G_i=S_{n/p_i}.$$

All embeddings are the standard ones just fixed.

Theorem 2. *If $d>1$, the resulting rank-k coset pregeometry is residually weakly primitive. In fact every available one-step inclusion in its intersection lattice is maximal, and all its 2^k intersections are distinct.*

Proof. Repeated use of Lemma 1 gives, for every $J\subseteq I$,

$$G_J=S_{n_J},\qquad n_J=\frac{n}{\prod_{j\in J}p_j}.$$

For $i\notin J$ the ratio $n_J/n_{J\cup\{i\}}$ is the prime p_i. The smaller exponent is at least $d>1$. Classical subfield maximality therefore applies to every such inclusion. Unique factorization makes the integers n_J distinct for distinct subsets J. The order $2^{2r}(2^{2r}+1)(2^r-1)$ of S_r is strictly increasing with r, so the corresponding subgroups are distinct as well. These statements give every condition in the source's definition. □

For an explicit example take

$$n=3^2\cdot5\cdot7\cdot11=3465,\qquad(p_1,p_2,p_3,p_4)=(3,5,7,11).$$

The four maximal parabolics are

$$S_{1155},\quad S_{693},\quad S_{495},\quad S_{315},$$

and their common intersection is $S_3=\mathrm{Sz}(8)$. Theorem 2 proves that this is a rank-four residually weakly primitive pregeometry.

Theorem 3. *The displayed pregeometry is absent from Constructions 1–11 in the fixed published edition of* The residually weakly primitive pre-geometries of the Suzuki simple groups. *It therefore contradicts the literal exhaustive assertion in its Theorem 5.*

Proof. Constructions 1–10 all include a soluble maximal parabolic: a dihedral group, an affine soluble group, a cyclic group extended by a group of order four, or the specified mixed root group. Every maximal parabolic in our example is nonabelian simple. Thus a permutation of types and an ambient group automorphism cannot identify our system with any of those constructions.

Construction 11 writes $n=p_1^{e_1}\cdots p_m^{e_m}$ with the p_i pairwise distinct, and permits ranks $\alpha\in\{2,\dots,m-1\}$. Here $m=4$, so its largest permitted rank is three. Our rank is four. Theorem 5 states that all residually weakly primitive pregeometries of rank at least four are those in Constructions 1–11, with no additional operation of taking truncations. The example is therefore outside that literal list. □

3 Attribution and limits

The source is D. Leemans, *The residually weakly primitive pre-geometries of the Suzuki simple groups*, Note di Matematica 20(1) (2000/2001), 1–20. The relevant definition is on printed pages 4–5, the construction conventions on page 15, and Construction 11 and Theorem 5 on page 18. The source was received on 26 July 1999 and accepted on 21 September 1999.

The argument is a reworking of Sections 2–3 of the project's earlier Suzuki subfield manuscript, whose separate review already checked this exact omitted boundary against the source. The old proof establishes additional geometry and flag-transitivity statements using descent. Those stronger conclusions are not needed here and are not acquired as premises of this note. The elementary matrix intersection argument comes from the earlier project treatment of compatible Suzuki tori. Classical Suzuki structure and prime-degree subfield maximality remain explicit retained inputs.

The correction concerns the literal rank range. The condition

$$n/(p_1 \cdots p_k) > 1$$

includes the example; suitable extensions of the construction can therefore repair this omission. We make no claim that such a repair exhausts all embeddings, and do not classify the remaining ten constructions. Neither a new mathematical discovery nor historical priority is claimed by this reworking. It contributes no additional conjecture solution, Lean theorem or compression factor. The supplementary control checks the matrix parameter identities symbolically, and the sixteen exponents and thirty-two prime-degree inclusions by integer arithmetic. It does not enumerate S_{3465}.

Compatible tori and a missing Suzuki geometry

Human Compression project

Abstract

A common binary matrix algebra places two nonsplit tori in compatible subfield groups. The resulting rank-three geometry has maximal parabolics Sz(8), Sz(32) and $C_{533} : C_4$, and is absent from the stated exhaustive classification. We rework the previously reviewed explicit example, replacing the finite intersection census by a coprime-order argument. A separate corollary of the scalar-subspace construction contradicts the same source's rank-two uniqueness assertion. Neither argument gives a corrected complete classification.

1 Statement and retained inputs

For subgroups G_i of G, the elements of the coset pregeometry are the labelled left cosets gG_i; incidence is nonempty intersection. Write $G_J = \bigcap_{j\in J} G_j$ and $G_\varnothing = G$. The definition on printed page 46 of [1] calls this system residually weakly primitive if for every proper subset J of the type set some unused i makes $G_{J\cup\{i\}}$ a maximal proper subgroup of G_J. We use exactly this definition.

Theorem 1. *In $G = \mathrm{Sz}(32768)$ there are standard subfield subgroups $S = \mathrm{Sz}(8)$, $T = \mathrm{Sz}(32)$ and a subgroup $K = C_{533} : C_4$ with*

$$S \cap T = C_5 : C_4, \quad S \cap K = C_{13} : C_4, \quad T \cap K = C_{41} : C_4, \quad S \cap T \cap K = C_4. \tag{1}$$

Their rank-three coset geometry is residually weakly primitive, thick, flag transitive and residually connected. Its intersection lattice is absent from the applicable cases of Theorem 5.5 of [1].

The standard faithful Suzuki matrices are denoted by $U(a, b)$, $h(t)$ and w, where w is the antidiagonal identity matrix. Their formulas, Bruhat forms and exact coefficient-field intersections are given in [2, Lemma 1]. In particular, for standard embeddings, $\mathrm{Sz}(2^a) \cap \mathrm{Sz}(2^b) = \mathrm{Sz}(2^{\gcd(a,b)})$, including the degree-one group $\mathrm{Sz}(2) = C_5 : C_4$. We retain the classical Suzuki order formula

$$|\mathrm{Sz}(q)| = q^2(q^2 + 1)(q - 1), \tag{2}$$

maximality of simple subfield groups of prime relative field degree, and maximality of the full normalizers of nonsplit cyclic tori of orders $q \pm \sqrt{2q} + 1$. Such a normalizer has order four times the torus order. These structural facts are among the inputs used by [1, Section 3]; completeness of its geometry lists is not used here. Our common proof [2, Lemma 1], rather than a new subfield-intersection calculation, is an explicit prerequisite.

2 One algebra, two compatible tori

All displayed integer entries in this section encode binary polynomial coefficients in

$$F = \mathbb{F}_2[z]/(z^{15} + z + 1).$$

Thus, for example, 5682 denotes $\sum_{i=0}^{14} \epsilon_i z^i$ with binary digits ϵ_i of 5682; it is not an integer reduced modulo 32768. The polynomial is irreducible. The supplied arithmetic control also checks this directly: successive multiplication by z visits all 32767 nonzero polynomial classes before returning to one.

Put

$$M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} = U(1,0).$$

Direct multiplication gives

$$M^4 + M + I = 0, \quad |M| = 15, \quad |r| = 4, \qquad r^{-1}Mr = M^2. \tag{3}$$

The polynomial $X^4 + X + 1$ has no binary root and is not divisible by $X^2 + X + 1$, the only irreducible quadratic over $\mathbb{F}_2$. Consequently $\mathbb{F}_2[M] \cong \mathbb{F}_{16}$. Its scalar extension to F is a field because $\gcd(4, 15) = 1$.

Take $\lambda_8 = 5682$ and $\lambda_{32} = 624$ in the stated encoding, and set

$$a = (M + \lambda_8 I)^{315}, \qquad b = (M + \lambda_{32} I)^{25575}. \tag{4}$$

Here $315 = (8^4 - 1)/13$ and $25575 = (32^4 - 1)/41$. The following table gives their full open Bruhat parameters:

g	u	v	t	x	y	$\lvert g\rvert$
a	893	0	5682	893	893	13
b	5992	4671	1131	5992	1831	41

$$g = U(u,v)h(t)wU(x,y). \tag{5}$$

Equations (3)–(5), the scalar identities $\lambda_8^8 = \lambda_8$, $\lambda_{32}^{32} = \lambda_{32}$, and the indicated orders are finite polynomial identities in F. They are checked by the short witness program supplied with this note, using repeated squaring and the displayed modulus. Each entry of the first parameter row is fixed by eighth powering; each entry of the second is fixed by thirty-second powering. Thus $a \in S$ and $b \in T$. The table is also a direct membership certificate, independently of any enumeration of a Suzuki group.

Since a and b are polynomials in M, they commute. Their distinct prime orders give $|ab| = 13 \cdot 41 = 533$ and $\langle ab \rangle = \langle a, b \rangle$. With $c = ab$, the remaining matrix identity is

$$r^{-1}cr = c^{278}, \qquad 278^2 \equiv -1 \pmod{533}. \tag{6}$$

It follows that $K = \langle c, r \rangle = C_{533} : C_4$: the two cyclic factors have coprime orders, and (6) is an automorphism of order four. Every element has a unique form $a^i b^j r^k$, with $0 \le i < 13$, $0 \le j < 41$ and $0 \le k < 4$. In particular $|K| = 2132$. This K is not the full normalizer of C_{533} in the ambient group; no such maximality is asserted.

Lemma 2. *The intersections are exactly those in (1).*

Proof. The standard-subfield result gives $S \cap T = \mathrm{Sz}(2)$ of order twenty. Both a and r belong to S. If $a^i b^j r^k \in S$, multiplication on the left by a^{-i} and on the right by r^{-k} gives $b^j \in S$. But 41 does not divide $|S| = 29120$. Since b has order 41, we must have $j = 0$. Hence $K \cap S = \langle a, r \rangle$, of order 52. Similarly 13 does not divide $|T| = 32537600$, so $K \cap T = \langle b, r \rangle$, of order 164. Their common intersection contains $\langle r \rangle$ and has order dividing $\gcd(52, 164) = 4$, proving the final equality. The groups of orders 52 and 164 are the full nonsplit torus normalizers in S and T: (6) normalizes their cyclic factors, and the retained normalizer order is respectively $4 \cdot 13$ and $4 \cdot 41$. □

3 The geometry and the source contradiction

We verify all seven instances of residual weak primitivity. The subgroup S is maximal in G because the relative field degree is $15/3 = 5$. Inside S and T, respectively, choose the maximal torus normalizers $S \cap K$ and $T \cap K$. Inside K, choose $K \cap S$, of prime index 41. At each pairwise intersection the final subgroup $\langle r \rangle$ has prime index, respectively 5, 13 and 41. These are all proper type subsets. All eight intersections are distinct, by their orders. This verifies the exact subgroup condition of the source.

For clarity, the geometry properties can be proved without its general rank-three assertion. We use the following elementary observation. If A, B, C are subgroups and

$$A = (A \cap B)(A \cap C), \tag{7}$$

every triple flag of their left-coset system has a common point. Translate the A-coset to A, and write the other two cosets as bB, cC with $b, c \in A$. Factor $b^{-1}c = xy$ with $x \in A \cap B$, $y \in A \cap C$. Then $bx = cy^{-1}$ lies in all three cosets. Every flag of fewer types also extends to a chamber: an incident pair has a common point, and that point supplies the missing coset. Thus (7) gives a geometry and transitivity on every type of flag.

In our example the product set $(K \cap S)(K \cap T)$ has cardinality

$$\frac{52 \cdot 164}{4} = 2132 = |K|.$$

It is contained in K, so equals K. No assumption that an arbitrary subgroup product is a subgroup is needed. Apply (7) with $A = K$. The three panel sizes are 5, 13 and 41, so the geometry is thick.

For residual connectivity, the two residue subgroups in S generate S: one is its maximal torus normalizer of order 52, while the other, $S \cap T$, contains an element of order five and is not contained in that normalizer. The same argument, using order 164, works in T. The product formula gives generation in K. Finally $\langle S, T \rangle = G$, because S is maximal and 31 divides $|T|$ but not $|S|$. The usual coset-graph argument now proves connectedness of the whole incidence graph and of every rank-two residue: a component containing the base flag consists exactly of the cosets reached by words in its incident stabilizer subgroups. Those subgroups generate the relevant group, as just shown. Flag transitivity transports this statement to all residues.

It remains to compare the actual source conclusion. Here the odd field degree is $15 = 3 \cdot 5$, whose prime exponents sum to two. Moreover the subfield $S = \mathrm{Sz}(8)$ has $8 - 1$ prime. The applicable portion of [1, Theorem 5.5, printed pp.58–59] is therefore its first five shapes. Only shape 5 has two Suzuki maximal parabolics and one odd cyclic group extended by C_4. It prescribes the intersection of the two Suzuki parabolics as $C_4 \times C_2$, of order eight; our intersection has order twenty. A permutation of types cannot change this discrepancy, since K is soluble and S, T are not. Shape 8 belongs to the separate branch where the sum of prime exponents exceeds two, and cannot be used in this branch. This proves Theorem 1.

The source's closing qualification on page 61 says that its lists are necessary subgroup lattices, without establishing realizability of every listed shape. Our example contradicts necessity, so that qualification does not remove the counterexample. We do not claim the geometric intersection property. Nor is this rank-three example locally doubly transitive: in the local action on a panel of size thirteen, the stabilizer of one point has order four and cannot be transitive on the other twelve points.

4 A separate rank-two consequence

The preceding example should be distinguished from the locally doubly transitive geometries in [3].

Corollary 3. *The uniqueness assertion for composite $q-1$ in [1, Theorem 6.1, printed p.61] is false at $q = 32768$.*

Proof. Here $q-1 = 32767 = 7 \cdot 31 \cdot 151$ is not prime. The source asserts that the unique residually weakly primitive and locally doubly transitive pregeometry has rank two and parabolics $D_{2(q-1)}$ and $q : (q-1)$, with common subgroup of order $q-1$. Its two valencies are 2 and q.

Apply [3, Theorem 3] to $F = \mathbb{F}_{32768}$ and $K = \mathbb{F}_8$. For each two-dimensional K-subspace T containing K, it gives a rank-two geometry with parabolics $\mathrm{Sz}(8)$ and Q_T, where $|Q_T| = 3584$ and their intersection has order 448. It is residually weakly primitive and locally doubly transitive, with valencies 65 and 8. These differ from 2 and 32768, even after interchanging types. Thus even the underlying incidence geometry is absent from the uniqueness assertion. The forty-one classes counted in [3, Corollary 5] use that paper's stated equivalence under automorphisms of the specified group; a count of forty-one bare incidence isomorphism classes is not needed or asserted here. □

5 Attribution and verification

The binary algebra, fixed scalars, matrices and rank-three construction were obtained in the project's earlier manuscript on compatible Suzuki tori, and were separately reviewed there. We retain that proof, its certificate, original program and review as provenance. The present note reuses the exact witness; its intersection proof uses group orders in place of the earlier enumeration of 2132 elements. The short new replay reads the fixed witness, verifies (3)–(6) and the arithmetic used in Lemma 2, and never searches for M or enumerates an ambient group. Classical Suzuki structure is retained explicitly.

The exact subfield-intersection lemma and the scalar-subspace theorem are acquired from [2, 3]; their complete proofs and reviews accompany this note. The rank-two corollary is a direct use of the latter theorem, not a separate discovery. The earlier higher-rank extension is not a premise. No other source cases or future realization questions are declared settled, and no complete corrected classification, publication priority, new conjecture solution or full Lean proof is claimed.

References

[1] D. Leemans, *The rank 3 geometries of the simple Suzuki groups* $\mathrm{Sz}(q)$, Note di Matematica 19(1) (1999), 43–63.

[2] Human Compression project, *A missing boundary in the Suzuki subfield construction*, reviewed project manuscript, 2026, Lemma 1.

[3] Human Compression project, *Scalar subspaces in rank-two Suzuki geometries*, reviewed project manuscript, 2026, Theorem 3 and Corollary 5.

Scalar subspaces in rank-two Suzuki geometries

Human Compression project

Abstract

A maximal subfield need not be maximal as an invariant additive subspace. This distinction changes a family in the rank-two Suzuki classification. We give the exact overgroups in the relevant mixed parabolic, construct the resulting geometries, and determine their equivalence by field automorphisms. For Sz(32768) this single family has forty-one isomorphism classes in the source's convention, whereas its table gives fourteen classes in total. The argument reuses the scalar-center correspondence already established in this project's work on higher-rank Suzuki configurations. It does not classify the remaining families.

1 An elementary overgroup correspondence

We first state the reusable group-theoretic fact. All groups in this note are finite. An action is called primitive when it is transitive and has no nontrivial invariant partition.

Lemma 1. *Let V be an abelian normal subgroup of Q, let $H \le Q$, and suppose $Q = VH$. Put $W = V \cap H$. The overgroups of H in Q correspond bijectively to the H-invariant subgroups of V containing W, by*

$$L \longmapsto L \cap V, \qquad T \longmapsto TH.$$

The coset action on Q/H is primitive precisely when V/W has no proper nonzero H-invariant subgroup. It is doubly transitive precisely when H is transitive on the nonzero elements of V/W.

Proof. If $H \le L \le Q$, write an element of L as vh. Since $h \in L$, we have $v \in L \cap V$, and therefore $L = (L \cap V)H$. Conversely, an H-invariant subgroup $T \le V$ makes TH a subgroup; because $W \le T$, its intersection with V is T. These operations are inverse. The usual correspondence between blocks containing the base coset and overgroups of its stabilizer proves the assertion about primitivity. Finally, V/W acts regularly on the cosets, identified with its additive set, and the base stabilizer acts by conjugation. Its transitivity away from zero is exactly double transitivity. □

If V is a vector space over a field K and the image of H on V is the full group of nonzero scalar multiplications, the invariant additive subgroups are exactly the K-subspaces. Thus an overgroup TH covers H precisely when $\dim_K(T/W) = 1$. On those cosets its permutation group is the affine group of the line over K. This is the scalar-center argument from the earlier project manuscript; the statement above isolates the group-theoretic reason for it.

2 Application to the Suzuki root group

Let $n > d \geq 3$ be odd integers, with $d \mid n$, and put $F = \mathbb{F}_{2^n}$, $K = \mathbb{F}_{2^d}$, $q = 2^n$, $s = 2^d$ and $\theta = 2^{(n+1)/2}$. We use the standard faithful four-dimensional Suzuki matrices, with

$$U(a,b) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ a & 1 & 0 & 0 \\ a^{\theta+1} + b & a^\theta & 1 & 0 \\ a^{\theta+2} + ab + b^\theta & b & a & 1 \end{pmatrix}.$$

Multiplication gives the root-group law

$$U(a,b)U(c,e) = U(a+c, b+e+ac^\theta), \qquad Z_F = \{U(0,b) : b \in F\}.$$

The diagonal torus consists of

$$h(t) = \mathrm{diag}(t^{\theta+1}, t, t^{-1}, t^{-\theta-1}), \qquad h(t)^{-1}U(a,b)h(t) = U(t^\theta a, t^{\theta+2}b). \tag{1}$$

Let U_K be given by $a, b \in K$, let $H_K = \{h(t) : t \in K^*\}$, and put $B = U_K H_K$. These are the standard root group, torus and point stabilizer in $S = \mathrm{Sz}(s) \leq G = \mathrm{Sz}(q)$. The ambient exponent θ restricts to the Suzuki exponent on K: the two exponent indices differ by a multiple of d.

We retain the classical Suzuki facts that this matrix model is faithful, that B is the stabilizer in the doubly transitive action of S on $s^2 + 1$ ovoid points, that S is maximal in G when n/d is prime, and that $\mathrm{Aut}(G) = G \rtimes \langle \phi \rangle$ with $\phi : x \mapsto x^2$ of order n. These structural results are due to Suzuki and Tits, and are also the structural inputs to the source classification: see its Section 2.2 and the structural list in Section 4, which cite Suzuki's *On a class of doubly transitive groups* (Annals of Mathematics 75 (1962), 105–145), Tits and Lüneburg. We do not reprove these classical facts here.

Both exponents in (1) are relatively prime to $q - 1$. This is immediate for θ. For the other, an odd common divisor of $2^n - 1$ and $2^{(n-1)/2} + 1$ would give simultaneously $2^n = 1$ and $2^{n-1} = 1$ modulo that divisor, forcing it to be one. The action of H_K on Z_F is consequently the full group of K-scalar multiplications.

Theorem 2. *Put $Q = Z_F B$. The overgroups of B in Q are exactly*

$$Q_T = Z_T B \quad (K \leq T \leq F), \tag{2}$$

where T ranges over K-subspaces and $Z_T = \{U(0,t) : t \in T\}$. Their orders are $|Q_T| = |T|s(s-1)$. In particular, the groups covering B are exactly those with $\dim_K T = 2$, and their order is $s^3(s-1)$. For every $c \in F \setminus K$,

$$\langle B, U(0,c) \rangle = Z_{K+Kc} B. \tag{3}$$

Proof. Here Z_F is abelian and normal in Q, and $Z_F \cap B = Z_K$. The root group U_K centralizes Z_F; the diagonal action is the scalar action just proved. Lemma 1 gives (2), the order formula, and the maximality criterion. Conjugating $U(0,c)$ by H_K generates exactly the additive space Kc, and B already contains Z_K. This proves (3). □

The source's Lemma 6.9, printed page 112, asserts instead that (3) has order $qs(s-1)$ and contains B maximally because K is a maximal subfield. If $n/d = r$ is an odd prime, then $r \geq 3$. There are K-subspaces strictly between K and F, regardless of the absence of intermediate fields. Thus B is never maximal in this Q. This invalidates the corresponding mixed-parabolic row of Tables 8 and 11. The distinction is visible already at $s = 8$, $q = 512$:

$$|B| = 448 < 3584 < |Q| = 28672. \tag{4}$$

3 The actual geometries and their equivalence

Suppose from now on that $r = n/d$ is prime. For every two-dimensional K-subspace T containing K, form the rank-two coset geometry

$$\Gamma_T = \Gamma(G; S, Q_T). \tag{5}$$

As in the source, elements are left cosets and incidence means nonempty intersection. Isomorphism means an automorphism of the specified group G carrying one subgroup configuration to the other, with type interchange allowed. We make no assertion here about isomorphisms that do not preserve the specified group action.

Theorem 3. *Every geometry (5) is firm, residually connected, flag transitive, residually weakly primitive, and locally doubly transitive. Its two local degrees are $s^2 + 1$ and s. Two members are conjugate under G precisely when their spaces T are equal. Their isomorphism classes in the source's convention are exactly the orbits of ϕ on*

$$\mathbb{P}_K(F/K). \tag{6}$$

There are $(s^{r-1} - 1)/(s - 1)$ conjugacy classes before this field fusion.

Proof. Coordinates in the standard subfield matrices give $S \cap Q_T = B$. More explicitly, every element of Q_T has a unique expression $U(a, b)h(t)$ with $a \in K$, $b \in T$, $t \in K^*$; such an element is in S exactly when $b \in K$. The subgroup S is maximal in G. Since $T \neq K$, the group Q_T is not contained in S, and therefore $\langle S, Q_T \rangle = G$. The incidence graph is consequently connected. A rank-two coset system is a geometry and is flag transitive: every element has an incident partner, and each incident pair is a translate of the base pair. Its two valencies are $[S : B] = s^2 + 1$ and $[Q_T : B] = s$, so it is firm. In rank two, connectedness supplies all the residual connectivity conditions of rank at least two.

The stabilizer B is maximal in S by the classical primitive ovoid action, and is maximal in Q_T by Theorem 2. Along with maximality of S in G, these are exactly the source's rank-two residual weak primitivity conditions. The action of S on its local cosets is doubly transitive. By Lemma 1, the other local action is the affine action of $K \rtimes K^*$ on K, also doubly transitive. This proves the first assertions.

The group S is nonsoluble whereas Q_T is soluble, so an isomorphism cannot exchange their types. The joint normalizer of S and B in G is B: maximality and simplicity give $N_G(S) = S$, and B is self-normalizing in S because it fixes a unique ovoid point. Every element of B preserves each T, by the scalar action. Hence no two different spaces in (2) give conjugate configurations.

Both S and B are preserved by ϕ. Their joint normalizer in $\mathrm{Aut}(G)$ is thus $B \rtimes \langle \phi \rangle$. After the trivial B action is removed, equivalence is exactly field fusion of the spaces T. Such spaces are the inverse images of lines in F/K, giving (6) and its stated cardinality. □

4 Two exact field counts

The following counts use the full automorphism group of G, not just inner conjugacy. They require no Suzuki group census.

Lemma 4. *Let σ be an odd-order field automorphism of F preserving K. Every σ-fixed element of F/K has a representative in F^σ. Consequently*

$$(F/K)^\sigma = F^\sigma/K^\sigma.$$

Proof. If σ has odd order a and the class of v is fixed, the sum $v + \sigma(v) + \cdots + \sigma^{a-1}(v)$ is a fixed representative of that same class: its image in F/K is $av = v$ in characteristic two. The kernel of the map from fixed representatives is K^σ. □

First take $K = \mathbb{F}_8$, $F = \mathbb{F}_{512}$. There are nine lines in F/K. The map ϕ^3 is K-linear of order three. An invariant line would have an eigenvalue in K^* of order dividing three, hence eigenvalue one. But its fixed space is zero by Lemma 4, since $F^{\phi^3} = K$. Therefore ϕ^3 fixes no line. The orbits of the order-nine map ϕ can only have lengths one, three or nine; the first two would be fixed by ϕ^3. All nine lines form one orbit.

Now take $K = \mathbb{F}_8$, $F = \mathbb{F}_{32768}$. There are

$$(8^4 - 1)/7 = 585 \tag{7}$$

lines in F/K. The K-linear map ϕ^3 has order five and no invariant line: the same argument uses $\gcd(5, 7) = 1$ and $F^{\phi^3} = K$. Hence any power ϕ^j with $5 \nmid j$ fixes no line, because its generated group contains ϕ^3.

Put $\sigma = \phi^5$, of order three. Its fixed field is $\mathbb{F}_{32}$, and its fixed field on K is $\mathbb{F}_2$. Thus Lemma 4 gives a fixed subspace of F/K with sixteen elements. On K, σ acts as the fourth power. If a K-line is invariant, choose a nonzero vector v on it and write $\sigma(v) = \lambda v$. There is a unique $k \in K^*$ with $k^3 = \lambda^{-1}$, because $\gcd(3, 7) = 1$. Then kv is fixed. Conversely a fixed nonzero vector spans an invariant line. Each such line contains exactly one nonzero fixed vector, since the fixed scalar field is $\mathbb{F}_2$. There are therefore fifteen fixed lines for each of ϕ^5 and ϕ^{10}. Burnside's orbit count now gives

$$\frac{585 + 15 + 15}{15} = 41. \tag{8}$$

There are three orbits of length five and thirty-eight of length fifteen.

Corollary 5. *For $G = \mathrm{Sz}(32768)$, the geometries (5) with $S = \mathrm{Sz}(8)$ give forty-one isomorphism classes in the source's convention. All are locally doubly transitive. Theorem 7.1 and its Table 11 omit this family, and the total fourteen in Table 12 at $e = 7$ is false.*

Proof. The count is (8). The group Q_T has order 3584 and a nonabelian normal Sylow two-subgroup. None of the other rows with the nonsoluble parabolic Sz(8) has this subgroup type: their other parabolic is Suzuki, dihedral, a cyclic group extended by a group of order four, or the mixed group of order $32768 \cdot 8 \cdot 7$. Thus (5) is absent from the asserted exhaustive list. Forty-one classes in one family also contradict the printed total fourteen. The same order argument already gives an omitted family at $q = 512$. □

Attribution and scope. The target is D. Leemans, *The Rank 2 Geometries of the Simple Suzuki Groups* Sz(q), Beiträge zur Algebra und Geometrie 39(1) (1998), 97–120: Lemma 6.9, Theorem 7.1 and Tables 8, 11, 12. The source's classification is considered under its specified-action equivalence, defined on printed page 100. The matrix model, ovoid action, subgroup maximality and ambient automorphism description are retained classical inputs. The scalar-center argument was already proved and separately reviewed in the project's earlier Suzuki subfield manuscript, Section 4; the new contribution here is its application to this rank-two row, the genuine replacement family, its local double transitivity and its equivalence calculation. No historical priority is asserted. The remaining rows and the complete corrected global count are not established by this note. This corrects a published classification; it is not an answer to a named open conjecture.

The supplementary program independently constructs the fields of orders 512 and 32768, verifies their defining polynomials by Rabin's criterion, lists all two-dimensional $\mathbb{F}_8$-subspaces containing $\mathbb{F}_8$, and checks every field orbit. It agrees with (7)–(8). These finite controls are not premises of the written proof.